

Lecture 24

Implementing Object Features without Meta Features

Group Work

Add each of the following operators to Trefoil as a **Trefoil function** (if possible) or as a language **primitive** (if needed)

1. Power (you can assume exponent is positive)

- $(\text{pow } 2 \ 3) \Rightarrow 8$
- $(\text{pow } 4 \ 2) \Rightarrow 16$

2. Absolute value

- $(\text{abs } 5) \Rightarrow 5$
- $(\text{abs } -6) \Rightarrow 6$

Group Work

Implement options in Trefoil 2 ways:

1. Using OCaml options

```
type value =
```

```
  ...  
  | Option of value option
```

2. Not using OCaml options

```
type value =
```

```
  ...  
  | Option of <SOMETHING>
```

```
type value =
```

```
  ...  
  | None  
  | Some of <SOMETHING>
```

```
none  
(some e)  
(get e)  
(is-some e)
```